

WILLOW GROVE NAS, PA, USA

WMO: 724086

Lat: 40.200N

Lon: 75.150W

Elev: 110

StdP: 100.01

Time Zone: -5.00 (NAE)

Period: 90-11

WBAN: 14793

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-8.6	-20.4	0.6	-8.6	-18.2	0.8	-6.5	10.6	-0.3	9.1	0.5	3.3	300	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	33.3	23.3	31.9	22.6	30.5	21.9	25.1	30.9	24.2	29.4	23.4	28.3	2.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.3	28.1	22.5	17.4	27.1	21.8	16.7	26.4	76.7	30.7	73.0	29.5	69.8	28.2	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.5	6.6	-15.3	35.7	2.3	1.4	-17.0	36.7	-18.3	37.6	-19.6	38.4	-21.2	39.4		
(5)			-16.0	27.0	2.1	1.4	-17.5	28.0	-18.7	28.8	-19.9	29.5	-21.4	30.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.4	0.6	1.5	5.8	11.6	16.7	22.1	24.4	23.2	19.0	13.1	7.3	2.8		
	DBStd	9.45	5.42	5.26	5.18	4.99	4.19	3.61	2.96	2.77	3.91	4.53	4.87	4.49		
	HDD10.0	1082	297	241	152	37	1	0	0	0	0	18	112	226		
	HDD18.3	2742	551	471	391	210	83	9	1	2	39	172	332	480		
	CDD10.0	1958	4	3	21	86	211	362	446	409	269	115	29	4		
	CDD18.3	574	0	0	1	9	34	121	188	153	58	11	0	0		
	CDH23.3	5166	0	0	19	107	366	1124	1759	1298	426	63	3	0		
	CDH26.7	1798	0	0	3	36	114	399	696	438	106	7	0	0		
(14) Wind	WSAvg	2.6	3.1	3.5	3.4	3.3	2.6	2.3	2.1	1.9	2.1	2.2	2.7	3.0		
(15) Precipitation	PrecAvg	1212	92	72	102	97	122	105	110	105	100	81	98	96		
	PrecMax	1752	348	172	214	247	241	355	302	246	234	178	274	213		
	PrecMin	778	18	21	41	18	34	13	26	27	36	30	13	17		
	PrecStd	230	63	32	43	46	61	76	65	53	55	34	66	60		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	18.7	26.3	30.9	32.4	34.4	36.0	33.9	31.7	27.5	22.9	17.7		
		MCWB	15.5	11.7	18.1	17.7	21.7	23.9	24.6	23.2	21.9	20.5	15.5	15.1		
	2%	DB	15.2	15.4	20.8	26.5	29.7	32.5	33.8	32.1	29.4	25.2	20.2	15.0		
		MCWB	12.7	10.7	14.0	16.3	19.7	23.0	23.6	22.7	21.3	19.0	15.0	12.7		
	5%	DB	11.3	12.3	17.0	23.1	27.4	30.6	32.1	30.7	27.6	23.0	17.8	12.1		
		MCWB	8.6	7.6	11.4	14.6	19.3	21.9	23.1	22.4	21.1	17.7	14.2	9.2		
	10%	DB	8.1	9.5	13.9	20.3	25.0	29.0	30.4	29.2	26.0	21.0	15.5	9.8		
		MCWB	5.4	5.8	9.4	13.3	17.0	20.8	22.2	21.5	19.9	16.5	11.9	6.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.4	14.7	19.0	19.7	24.2	25.7	26.6	25.6	24.4	21.8	18.3	15.8		
		MCDB	17.8	17.1	25.3	26.4	30.3	31.6	33.3	30.8	29.2	24.9	20.9	17.3		
	2%	WB	13.0	10.8	15.3	17.6	21.6	24.2	25.4	24.4	23.3	20.4	16.6	13.1		
		MCDB	15.2	14.7	19.1	24.5	27.6	30.2	31.5	29.6	27.2	24.0	18.5	14.8		
	5%	WB	8.9	8.4	12.4	16.0	20.0	23.2	24.4	23.6	22.2	19.1	14.4	9.9		
		MCDB	10.9	11.8	15.8	21.8	25.7	28.7	29.7	28.5	25.9	21.9	17.9	11.7		
	10%	WB	5.7	5.9	9.9	14.2	18.4	22.2	23.5	22.7	21.0	17.3	12.0	7.5		
		MCDB	7.7	9.4	13.6	19.2	23.4	27.1	28.4	27.3	24.5	20.5	15.2	9.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.8	9.7	11.0	11.3	10.8	10.2	10.4	10.3	10.9	9.2	7.9		
		MCDBR	11.3	12.6	14.2	14.4	14.5	12.6	11.9	11.7	11.5	12.2	12.1	11.3		
	5% WB	MCWBR	10.0	9.7	9.6	7.8	7.9	6.3	5.3	5.6	6.0	7.5	9.5	9.7		
		MCWBR	10.7	11.6	12.9	12.6	12.9	11.4	10.6	10.4	10.0	10.3	10.8	10.3		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.319	0.357	0.388	0.434	0.493	0.492	0.486	0.401	0.356	0.324	0.306		
	taud		2.460	2.413	2.375	2.317	2.215	2.090	2.108	2.111	2.321	2.457	2.512	2.514		
	Ebn at Noon		858	898	895	886	848	796	792	784	837	844	833	833		
	Edh at Noon		78	95	110	124	141	160	156	151	115	90	75	70		
(44) All-Sky Solar Radiation	RadAvg		1.89	2.79	3.69	4.79	5.33	5.83	5.96	5.17	4.29	3.04	2.15	1.63		
	RadStd		0.16	0.31	0.29	0.43	0.62	0.48	0.29	0.33	0.49	0.32	0.19	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (69 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+120	+68

Nomenclature: See separate page